

SHADOWWATCH

SOC Training Simulator

A full-stack **Security Operations Center (SOC) training platform** built from scratch with PHP, MySQL, and vanilla JavaScript. Analysts triage live-simulated alerts, manage incident tickets, analyze logs, and compete on a leaderboard — all in a dark cyberpunk-themed interface.

Language	Database	Frontend	Auth & Security
PHP 8.x	MySQL 8.x	HTML5 / CSS3 / JS	bcrypt + CSRF + PDO

Features

■ Alert Console

- Real-time alert queue sorted by severity (Critical to Low)
- Acknowledge and resolve with response actions (Block IP, Isolate Host, etc.)
- Points awarded per action and severity level
- Generate random alerts from attack scenarios

■ Incident Management

- Create and track incident tickets (PICERL lifecycle)
- Status workflow: Open → Investigating → Contained → Closed
- Assign incidents to analysts (Senior Analyst / Admin only)
- Closing an incident awards +50 bonus points

■ Log Analyzer

- Classify simulated logs: Malicious / Suspicious / Benign / Unknown
- Extract IOCs (IPs, domains, hashes) with regex
- Generate realistic log batches for training sessions
- Points for correct classifications

■ Threat Intelligence

- Built-in IOC database: IPs, domains, file hashes
- Live indicator lookup with confidence scoring
- 10 pre-loaded attack scenarios (Ransomware, APT, DDoS, Phishing...)
- Correlate intel findings with active alerts

■ Gamification & Ranking

- 10-tier rank system: Trainee → Elite Operator
- XP-based progression with visual level bar
- Badge system: First Blood, Alert Hunter, Log Master, etc.
- Live leaderboard with analyst stats

■ Admin Panel

- Role-based access: Analyst / Senior Analyst / Admin
- Manage users, scenarios, audit logs
- Bulk generate alerts and reset training sessions
- Secure: CSRF, bcrypt, prepared statements, input sanitization

Installation

Requirements

Requirement	Version
PHP	8.0+
MySQL	5.7+
Web Server	WAMP / XAMPP / MAMP

Step 1 — Clone the Repository

```
git clone https://github.com/yourusername/shadowwatch.git
```

Step 2 — Move to Web Server Directory

Environment	Path
WAMP	C:\wamp64\www\shadowwatch\
XAMPP	C:\xampp\htdocs\shadowwatch\
MAMP	/Applications/MAMP/htdocs/shadowwatch/

Step 3 — Create the Database

Open phpMyAdmin, click New, name it **shadowwatch**, click Create. Then go to the SQL tab and run the contents of **database/schema.sql**.

Step 4 — Configure Database Connection

Edit **includes/config.php**:

```
define('DB_HOST', 'localhost');
define('DB_USER', 'root');
define('DB_PASS', ''); // your MySQL password
define('DB_NAME', 'shadowwatch');
```

Step 5 — Open the App

```
http://localhost/shadowwatch/
```

Demo Accounts

All accounts use the password: **password**

Username	Role	Password
admin	Admin	password
jsmith	Senior Analyst	password
alee	Analyst	password
mchen	Analyst	password
rgarcia	Analyst	password

Scoring System

Action	Points
Acknowledge alert	+5
Resolve alert (base)	+25 to +50
Response action bonus	+10 to +30
Create incident	+25
Close incident	+50
Correct log analysis	+15
Partial log analysis	+5

Rank System

Level	Rank	XP Required
1	Trainee	0
2	Rookie	500
3	Analyst I	1,200
4	Analyst II	2,500
5	Senior Analyst	4,500
6	Threat Hunter	7,500
7	SOC Lead	11,000
8	Red Teamer	16,000
9	Cyber Specialist	22,000
10	Elite Operator	30,000

Project Structure

File / Folder	Description
admin/	Admin-only pages (users, scenarios, logs, settings)
api/	JSON API endpoints (alerts, incidents, logs, threat lookup)
assets/css/app.css	Full stylesheet — dark cyberpunk theme
assets/js/app.js	Frontend helpers (toasts, modals, API calls, CSRF)
database/schema.sql	Full DB schema + seed data
includes/auth.php	Authentication, roles, login/logout
includes/config.php	App configuration and DB constants
includes/db.php	PDO database singleton class
includes/functions.php	Helper functions (points, levels, alerts)
includes/header.php	Sidebar + topbar layout
includes/session.php	Session management and CSRF token handling
alerts.php	Alert console — triage and resolve
dashboard.php	Main SOC dashboard with stats
incidents.php	Incident ticket management
leaderboard.php	Analyst rankings and XP
logs.php	Log analyzer and IOC extractor
profile.php	Analyst profile and badges
scenarios.php	Threat intel and attack scenarios
settings.php	User account settings

Security Features

Security Feature	Implementation
CSRF Protection	All forms and API calls validated with session tokens
Password Hashing	bcrypt via PHP password_hash() — no plaintext storage
Prepared Statements	All DB queries use PDO with parameterized inputs
Input Sanitization	htmlspecialchars() applied to all user-generated output
Role-Based Access	Analyst / Senior Analyst / Admin — enforced server-side
Session Security	session_regenerate_id() on login, HTTPOnly cookies

How to Use

Step	Action
1	Login as admin and go to Admin → Settings
2	Click Generate Alerts (set count to 20+) to populate the queue
3	Go to Alert Console → click an alert → Acknowledge → Resolve
4	Pick a response action to earn points
5	Create Incident Tickets for complex or critical threats
6	Analyze logs in Log Analyzer for bonus points
7	Check your rank on the Leaderboard

Disclaimer: ShadowWatch is a training simulator. All threats, alerts, and incidents are entirely simulated. No real systems are monitored or affected.

Built as a cybersecurity portfolio project | MIT License